

The students decide not to use the bridge rectifier in the circuit and return to the original design of Figure 1. They next test the smoothing operation with a modified circuit shown in Figure 5 using  $C = 2000 \mu\text{F}$ , a switch,  $S$ , and a  $100 \Omega$  resistor. The switch is initially **open** so the resistor is disconnected at this time.

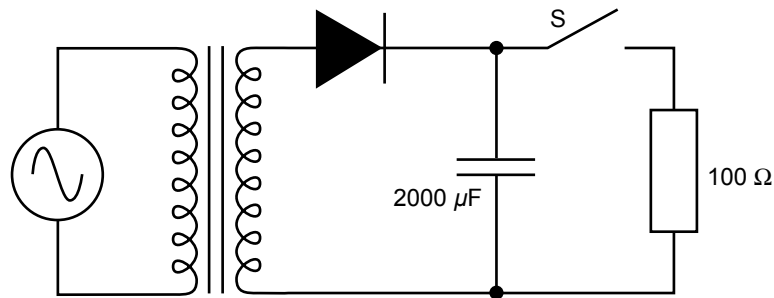


Figure 5

### Question 5

The time constant of the circuit shown in Figure 5 is measured by the students to be 2 ms. What is the effective forward resistance of the diode?

- A.  $0.1 \Omega$
- B.  $1 \Omega$
- C.  $2 \Omega$
- D.  $10 \Omega$

2 marks

The students now **close** the switch of the circuit of Figure 5, connecting the  $100 \Omega$  resistor into the circuit. The voltage across the  $100 \Omega$  resistor is measured using a CRO and is observed to comprise a DC component of about 12 V and an AC component, or ripple. They investigate this ripple by varying the values of some of the components in the circuit of Figure 5.

### Question 6

Which one of the following statements (A.–D.) concerning the magnitude of the ripple voltage is correct?

- A. It approximately doubles when the capacitance is halved to  $1000 \mu\text{F}$  and the resistance doubled to  $200 \Omega$ .
- B. It approximately doubles when the capacitance is doubled to  $4000 \mu\text{F}$  and the resistance halved to  $50 \Omega$ .
- C. It approximately doubles when the capacitance is halved to  $1000 \mu\text{F}$  and the resistance left unaltered at  $100 \Omega$ .
- D. It approximately doubles when the capacitance is doubled to  $4000 \mu\text{F}$  and the resistance left unaltered at  $100 \Omega$ .

2 marks

The students now consider the final part of the design, the voltage regulator, using a resistor,  $R_V$ , in series with a 9 V Zener diode (see Figure 1).

The current-voltage characteristic of the Zener diode is shown in Figure 6.

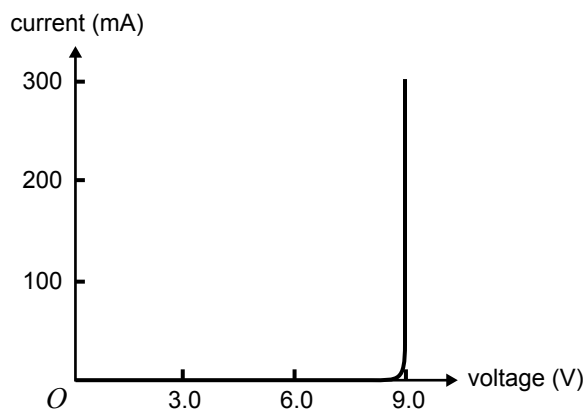


Figure 6

The students use a multimeter to measure the DC voltage across the  $30\ \Omega$  resistor in Figure 1 to be approximately 3 V.

### Question 7

Given that the voltage across the transformer is 24 V (peak-to-peak), in the space below explain why this value of 3 V across  $R_V$  is to be expected.

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2 marks

### Question 8

In this measurement for Question 7 the students could have used either a battery-powered multimeter (MM) or the mains-powered CRO.

Which one of the following statements concerning these two instruments is true?

- A. The MM (in DC voltage mode) is the most sensible choice to measure this voltage across  $R_V$  compared to a CRO as it is less likely to affect the power supply circuit.
- B. The CRO is the better voltage measuring instrument in this situation, as it can measure both the AC and DC voltages across  $R_V$ , while a MM can only measure DC voltages.
- C. The MM is the better instrument to use as the only way to measure a current through a resistor is by using a MM (in AC or DC current mode). When using a CRO there is no way that the current in the resistor can be determined.
- D. The CRO is the better instrument as it can be used to fully describe both the AC and DC voltages, while the MM cannot.

2 marks

**Question 9**

What is the power dissipated in the  $30\ \Omega$  resistor,  $R_V$ , of Figure 1? Express your answer in mW.

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|----|
| mW |
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2 marks

The students now have a fully operational regulated 9 V DC power supply as shown in Figure 1. The final circuit has  $C = 2000\ \mu\text{F}$ , a 9 V Zener diode regulator and  $R_V = 30\ \Omega$ . To start with, the students connect a load resistor of  $R_L = 300\ \Omega$ . They then do a series of tests to observe what might change the output DC across the load resistor.

**Question 10**

Based on their observations, which one of the following statements (**A.–D.**) is correct?

- A.** The output DC voltage changes if the mains supply voltage is increased to 250 V (RMS).
- B.** The output DC voltage changes if the load resistor is increased to  $400\ \Omega$ .
- C.** The output DC voltage changes if the 9 V Zener diode is replaced by a 6 V Zener diode.
- D.** The output DC voltage changes if the capacitor is increased to  $3000\ \mu\text{F}$ .

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2 marks

**Question 11**

The students have their circuit checked by a qualified electrician who advises them that their power supply circuit will require a heat sink. Explain why a heat sink is used in this regulated power supply circuit, and what part of the circuit needs to be attached to this heat sink.

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2 marks

As a final part of their project to design and construct a regulated DC power supply, the students have to write a brief report. Students discuss safety and risk associated with this project.

**Question 12**

Which is the most important activity to ensure maximum safety and minimum risk?

- A. learning from trial and error when designing and constructing electrical circuits
- B. getting the teacher (or someone appropriately qualified) to check each stage of the design and construction
- C. understanding how to correctly use a cathode ray oscilloscope and a multimeter for troubleshooting and diagnosis
- D. understanding the physics of specific components such as the transformer, diodes, capacitor and resistor

2 marks